• Security solutions: Tools and systems that helps SOC to protect an organization. They help SOC to detect attacks, analyze threats, Respond quickly, prevent damage.

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DEFENSIVE SECURITY INTRO:

SOC FUNDAMENTALS:

A SOC (Security operations centre) is a facility operated by a team of specialized security people. They protect an organization's network and resources by continously monitoring and identifying suspicious activities.
The main focus of SOC team is to keep DETECTION and RESPOND

• Detection:
a) Detect vulnerabilities
b) Detect unauthorized activity
c) Detect policy violations
d) Detect intrusions: Intrusions refers to unauthorized access to system and network

• Response: Response includes minimizing the effects of the thing that has been detected and perform a root cause analysis of the incident.

• The three pillars of SOC are People, Process and Technology which makes the SOC team ~~effectively~~ efficiently detect and response to a incident.

a) People: People in SOC will always be important. In a SOC with security solutions in place without people will end up focus on irrelevant issues. People helps the security solutions to truly identify harmful activities.

• Alert triage: It is a process of reviewing, validating and prioritizing security alerts to distinguish genuine threats from fake noise.

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CISO (~~...~~ chief information security officer)
↓
SOC Manager
↓

SOC analyst Level 1	SOC analyst Level 2	SOC analyst Level 3	Security Engineer	Detection Engineer

i) SOC analyst (Level 1): Any detection by security solutions would be first analyzed by them. They perform alert triage ~~and~~ to determine if a specific detection is harmful.

ii) SOC analyst (Level 2): ~~They do~~ Some detections requires deeper analysis so this analysis is done by SOC Level 2 after the Level 1 analysis is done.

iii) SOC analyst (Level 3): These analysts handles complex attacks, malware analysis, builds new detection rules (works like a detection engineer). If level 1 and 2 cannot fully understand an attack it comes to Level 3 analysts.

iv) Security enginers: These engineers deploys and configure security solutions to ensure their smooth operation.

v) Detection engineer: These engineers build the logic for the detection of harmful activities by security solutions.

vi) SOC manager: Manages the processes the SOC team follows and provides support.

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b) Process: Each ~~data~~ individual in a SOC team has its own Proccesses. The important processes involved in SOC are:

i) Alert triage: The first response to any alert is to perform triage. The triage is all about analyzing the alert and helps us prioritize it. It is about answering the 5Ws (what, when, where, who, why)

ii) Reporting: It means reporting the harmful alert to the higher analysts with the report containing the 5Ws answers along with a thorough analysis.

iii) Incident response and forensics: If the reported detection is very critical or harmful, high level team initiate a incident response. Sometimes forensics activity is also performed who aim is to incident's root cause.

c) Technology: The technology portion in the SOC pillar refers to security solutions. These tools minimize manual work of people. Some of the security solutions are:

i) SIEM (Security information and event ~~manager~~ management): It is a tool that collects logs from firewalls, EDR, servers, applications and detects suspicious patterns using the detection rules configured in SIEM solution and generate alert is any rule is ~~...~~ violated.

ii) EDR (Endpoint detection and response): EDR provides the SOC team with detailed real time and historical visibility of the device activity. It protects devices like laptops, Desktop, Server by monitoring processes, detecting malware and stopping Suspicious activity). So this way it performs detections and can also carry automated respond.

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Firewall: It is purely for network security and acts as a barrie-r for malicious or suspicious traffic. Firewall also has detection rules deployed.